

CROWN INSTITUTE

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FROM POSSIBILITY TO ACTUALITY

ICT313 Big Data for Software Development

LECTURE 11

Clustering

Preview

- What is Clustering
- Types of Clustering
- Exercise

What is Clustering

- Clustering is the task of :
  - dividing the population or data points into a number of groups
  - ***such that data points in the same groups are more similar to other data points in the same group than those in other groups.***
- In simple words, the aim is to segregate groups with similar traits and assign them into clusters.

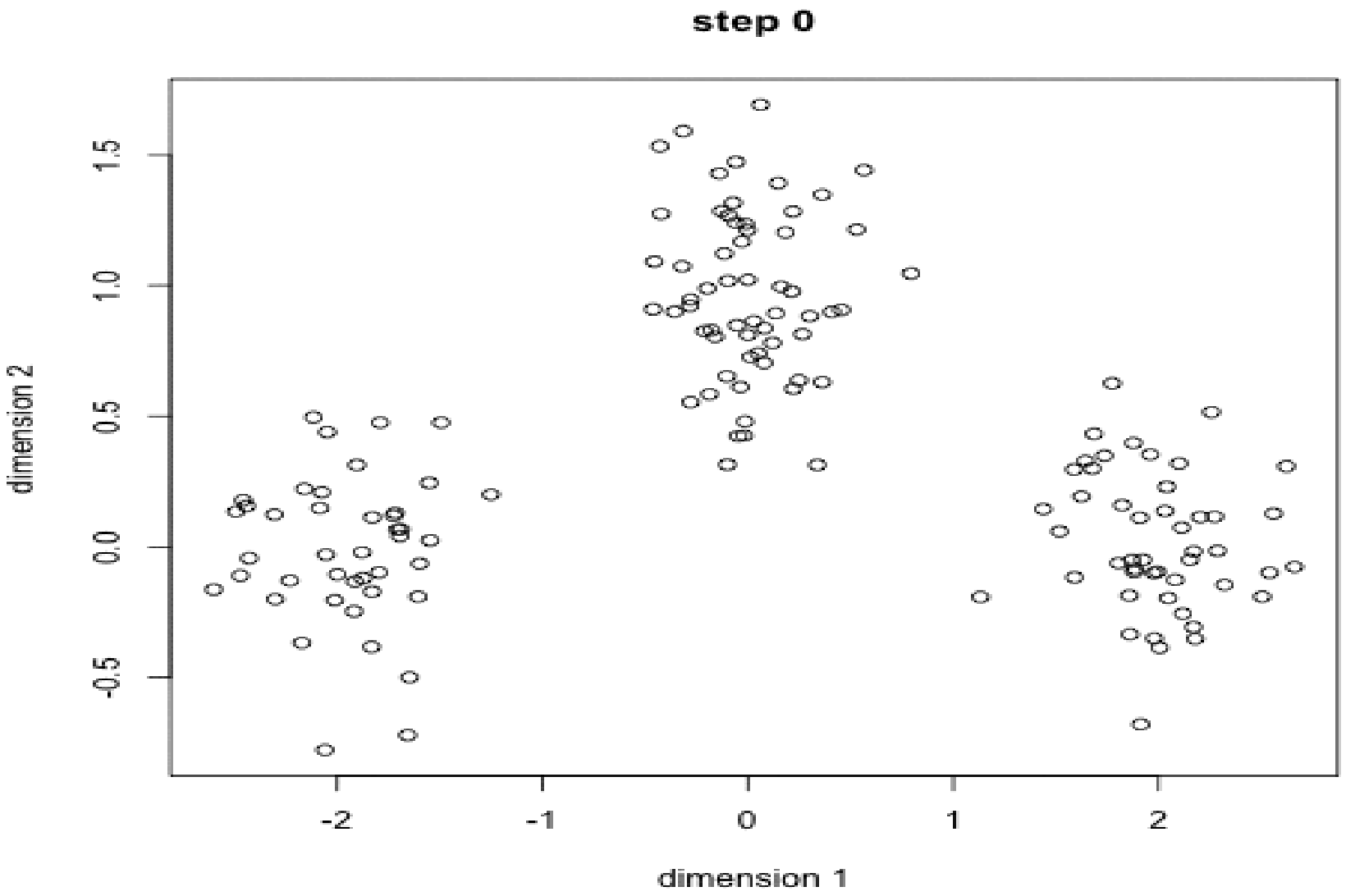
Types of Clustering

- **Hard Clustering:** In hard clustering, each data point either belongs to a cluster completely or not.
- **Soft Clustering**: In soft clustering, instead of putting each data point into a separate cluster, a probability or likelihood of that data point to be in those clusters is assigned

Types of Clustering Algorithms – Centroid Models

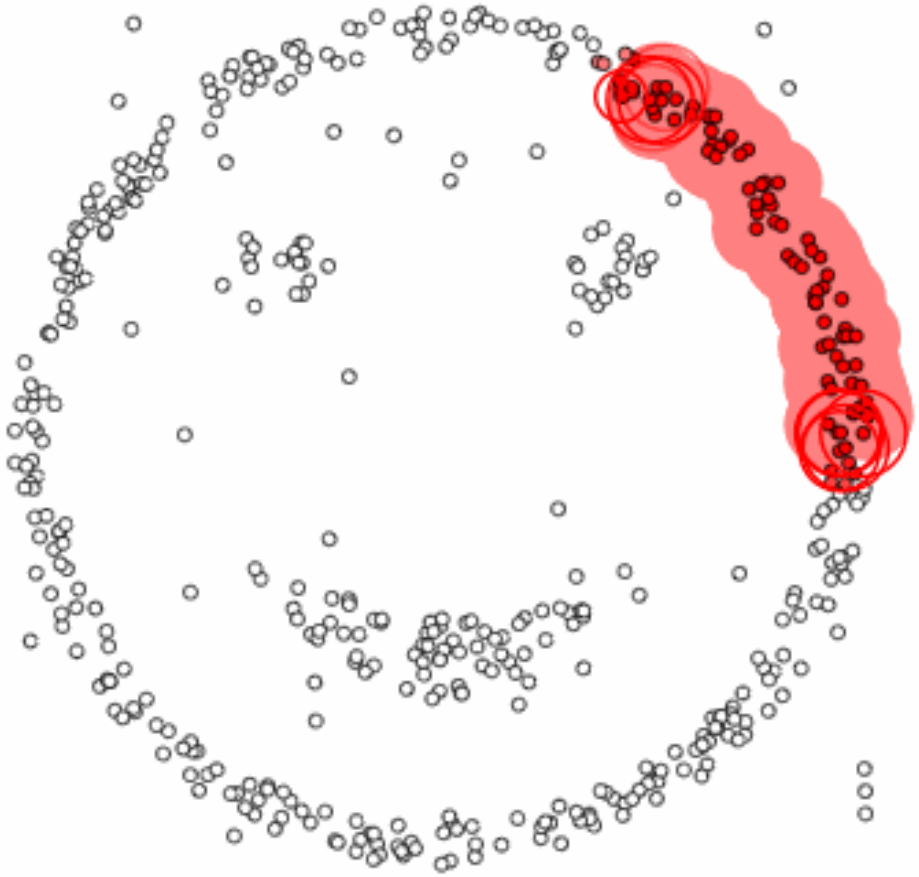
- Iterative clustering algorithms in which the notion of **similarity is derived by the closeness of a data point to the centroid of the clusters.**
- The no. of clusters required at the end have to be mentioned beforehand, which makes it important to have prior knowledge of the dataset.
- Run iteratively to find the local optima.
- **K-Means clustering algorithm** is a popular algorithm that falls into this category.

Types of Clustering Algorithms – Centroid Models



Types of Clustering Algorithms – Density Models

- Search the data space for areas of varied density of data points in the data space.
- It isolates various different density regions and assign the data points within these regions in the same cluster.
- Examples of density models are DBSCAN and OPTICS.



epsilon = 1.00
minPoints = 4

Restart

Pause

Types of Clustering Algorithms – Distributed Models

- Based on the notion of how probable is it that all data points in the cluster belong to the same distribution (For example: Normal, Gaussian).
- Often suffer from overfitting.
- Example of the models is Expectation-maximization algorithm which uses multivariate normal distributions.

Types of Clustering Algorithms – Connectivity Model

- These models are based on the notion that the ***data points closer in data space exhibit more similarity to each other*** than the data points lying farther away.

Types of Clustering Algorithms – Connectivity Models

- Two approaches.
  - Approach 1 - Start with classifying all data points into separate clusters & then aggregating them as the distance decreases.
  - Approach 2 - All data points are classified as a single cluster and then partitioned as the distance increases.

Types of Clustering Algorithms – Connectivity Models

- The choice of distance function is **subjective.** These models are very easy to interpret but lacks scalability for handling big datasets.
- Examples of these models are hierarchical clustering algorithm and its variants.

K Mean Clustering

- We first select a number of classes/groups to use and randomly initialize their respective center points.
- Each data point is classified by computing the distance between that point and each group center, and then classifying the point to be in the group whose center is closest to it.
- Based on these classified points, recompute the group center by taking the mean of all the vectors in the group.
- Repeat these steps for a set number of iterations or until the group centers don’t change much between iterations.

K Mean Clustering - Advantages

- Fast, as all we're really doing is computing the distances between points and group centers;
- Very few computations!
- Complexity is O(n * K * I * d),
  - where n: number of points, K: number of clusters, I: number of iterations, d: number of attributes

K Mean Clustering – Disadvantages

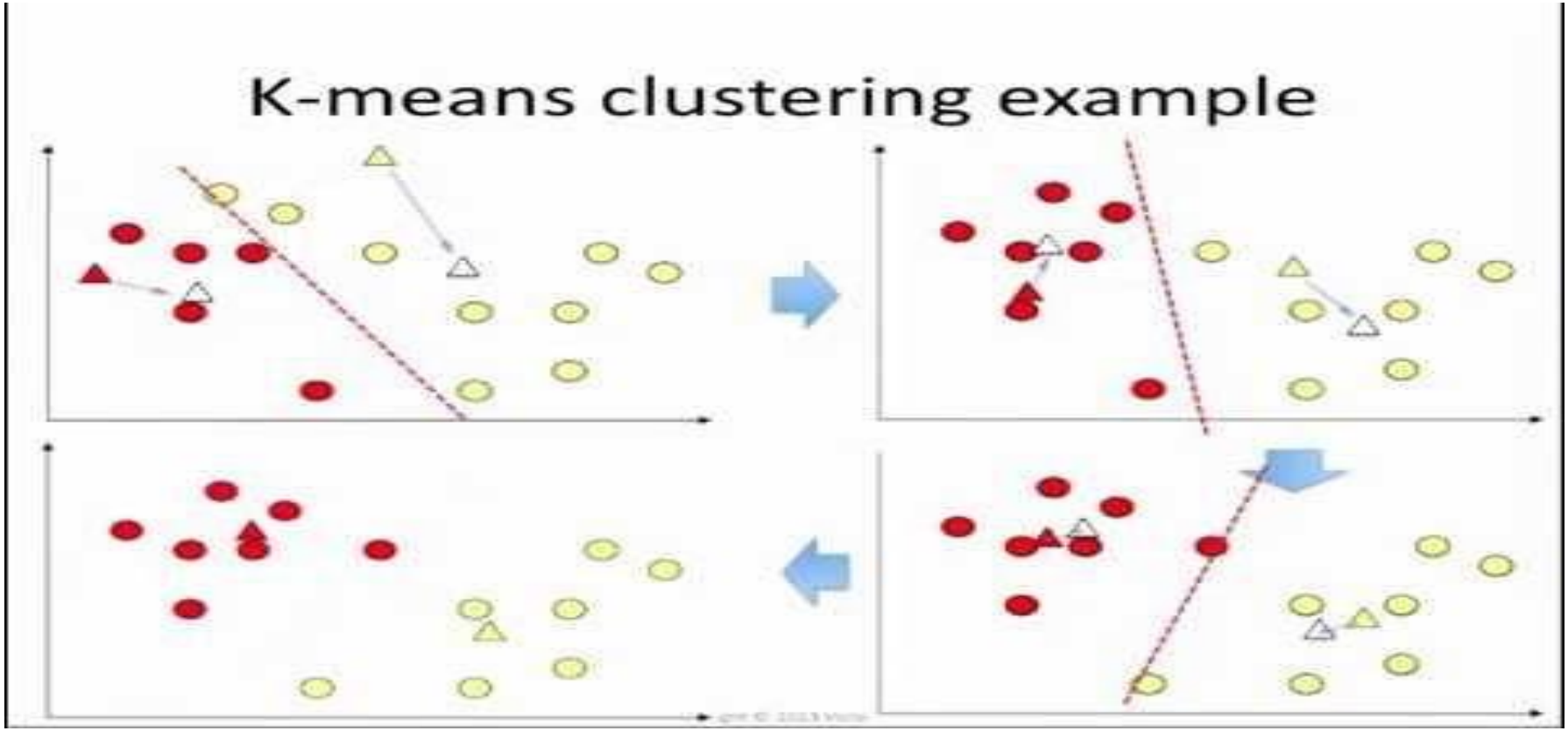
- Have to select how many groups/classes there are
- K-means also starts with a random choice of cluster centers and therefore it may yield different clustering results on different runs of the algorithm
- The results may not be repeatable and lack consistency

Evaluating K-means Clusters

- Most common measure is Sum of Squared Error (SSE)
  - For each point, the error is the distance to the nearest cluster
  - To get SSE, we square these errors and sum them.

$$SSE = \sum_{i=1}^{K} \sum_{x \in C_i} dist^2(m_i, x)$$

  - x is a data point in cluster C_i and m_i is the representative point for cluster C_i
    - can show that m_i corresponds to the center (mean) of the cluster
  - Given two clusters, we can choose the one with the smallest error
  - One easy way to reduce SSE is to increase K, the number of clusters
    - A good clustering with smaller K can have a lower SSE than a poor clustering with higher K



https://youtu.be/_aWzGGNrcic

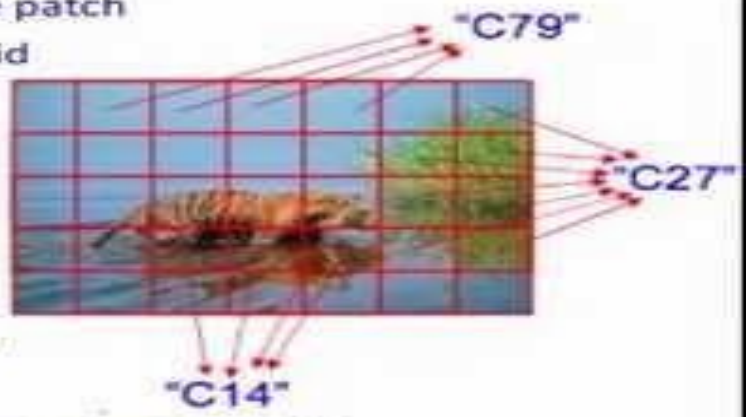
Application

- Image Interpretation and Pixel cleaning (PhotoShop)
- Segmentation of spending behavior
- As a precursor to Classification

K-means for image representation

- What K-means does:
  - groups all feature vectors from all images into K clusters
  - provides a cluster id for every patch in every image
    - represents the salient properties of the patch
    - similar-looking patches have the same id
- Represent patch with cluster id
  - image = bag of cluster ids
    - one for each patch in the image
  - K-dimensional representation:

{ 4 x "C14", 7 x "C27", 24 x "C79", 0 x everything else }



- similar to bag-of-words
- cluster ids sometimes called vis-terms or "visual words"

https://youtu.be/yDi2uX5tihc

Brain Tumor Segmentation using K-means Algorithm

https://youtu.be/0Ns7dRkwYjk?t=209